ARANYA CHATTERJEE

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EXPERIENCE

Warewolf | Machine Learning Intern *May 2026 – Present | Remote*

- Built and refined a YOLO-based object detection pipeline for surveillance footage, collaborating with a cross-functional engineering team throughout.
- Annotated ~780 surveillance images using Roboflow (bounding-box labeling, class organization); performed data cleaning/preprocessing (filtering, deduplication, normalization) and used OpenCV for frame extraction, preprocessing, and detection-output visualization across the training cycle.

EDUCATION

Madhav Institute of Technology and Science, Gwalior *2023 – 2027*
B.Tech — Computer Science Engineering | CGPA: 7.7

St. Jude's School, Shahdol — CBSE (Class XII), 86% *2022*
D.A.V Burhar, Burhar — CBSE (Class X), 91% *2020*

PROJECTS

Agentic AI Orchestration Engine — Google ADK & LangGraph *Aug 2026*
Tools: Google ADK, LangGraph, LangChain, Python, LiteLLM/Groq, DSPy, GEPA, Pydantic, PostgreSQL, FastAPI, Docker | GitHub

- Built a multi-agent GitHub-triage system on Google ADK — LLM-driven orchestration, LoopAgent batch workflows, Pydantic output verification, and bounded-retry-then-human-escalation logic.
- Reimplemented the same system in LangGraph as an explicit state graph with conditional edges, interrupt()/Command(resume=...) HITL, per-conversation thread isolation, and Postgres-backed checkpointing for crash-safe persistence.
- Bridged DSPy/GEPA reflective prompt optimization into both pipelines — GEPA-evolved instructions replaced hand-written prompts across the classification pipeline, re-optimized per target model.
- Containerized and deployed both implementations via FastAPI + Docker with Cloud Run-ready configuration.

MultimodalRAG — Intelligent Video & Document Assistant *January 2026*
Tools: Python, LangChain, Whisper AI, FAISS, Groq API, MoviePy, Streamlit, Sentence Transformers | GitHub

- Engineered a multimodal RAG system processing video (MP4, MOV, AVI) and documents (PDF, TXT, DOCX, CSV) via Whisper AI transcription and FAISS semantic search, achieving sub-second cross-modal retrieval with 95% retrieval accuracy and 40% faster video processing for files up to 100MB.
- Built a Streamlit interface with dark-theme UI and real-time AI-powered Q&A with precise source citations.

"MASTERJI" — RAG-Powered Chatbot *November 2025*
Tools: Python, LangChain, Hugging Face, FAISS, Llama-2, Streamlit | GitHub

- Architected a RAG system using LangChain and FAISS vector embeddings, achieving 30% improvement in response accuracy over base LLM outputs with sub-second query response and reduced hallucinations.
- Developed a Streamlit interface supporting multi-format uploads (PDF/TXT) with real-time Q&A, fully sourced citations, and context-aware generation via Llama-2.

TECHNICAL SKILLS

Programming: Python (Advanced), Java (HackerRank 5-Star), SQL (MySQL), C/C++

AI/ML & Computer Vision: PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, Hugging Face, YOLO, OpenCV, Roboflow, LLM Fine-Tuning, DSPy, GEPA, Google ADK, LangGraph, LiteLLM

Data Science & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Exploratory Data Analysis (EDA)

Tools & DevOps: Git, GitHub, FastAPI, Docker, PostgreSQL, VS Code, Jupyter Notebook

LEADERSHIP & POSITIONS OF RESPONSIBILITY

Team Leader — Code-N-Compete National Hackathon (IIT Kanpur) *April 2025*

- Spearheaded a developer team in a high-pressure national hackathon, delivering a functional prototype within a 48-hour sprint and coordinating cross-functional workflows for a 25% faster integration phase.

Second Author — Research Publication (ICMCTR 2025) *May 2025*

- Co-authored "A Hybrid IoT + AI Approach for Risk Management in Algorithmic Trading," presented at ICMCTR-2025, contributing to technical drafting and experimentation on risk mitigation for high-frequency financial systems.

CERTIFICATIONS & ACHIEVEMENTS

- National Finalist — Hack-O-Calypse (IIT Jammu) and Code-N-Compete (IIT Kanpur); published research at ICMCTR-2025.